

At the far end, there is the cyclic redundancy check (CRC) that detects cases of transmitted data being unintentionally changed or corrupted due to collisions. It is performed by comparing the CRC calculated by the transmitter with the CRC calculated by the receiver. If they are not the same, then corruption must have occurred, and the packet is discarded.

The message integrity code (MIC), also called a message authentication code (MAC), is a security feature that aids

the detection of data deliberately tampered with (as opposed to accidentally through collisions).

Error correction

There is a variant of the stack's physical layer (PHY) called the LE Coded PHY, which allows Bluetooth to not only detect errors but also correct them in packets up to certain limits. This aids communication at a low signal to noise ratio (SNR).

The further from the transmitter you go, the lower the signal strength and worse the SNR will become, making it

harder to keep the BER low. But with forward error correction (FEC), you can work at a low SNR and therefore things continue to be reliable at a much longer range.

Spread spectrum

Bluetooth addresses reliability challenges through the various ways in which it employs spread-spectrum techniques. So, rather than using a single channel for all packets, they get transmitted across a set of different channels, thus minimising any likelihood of collisions.

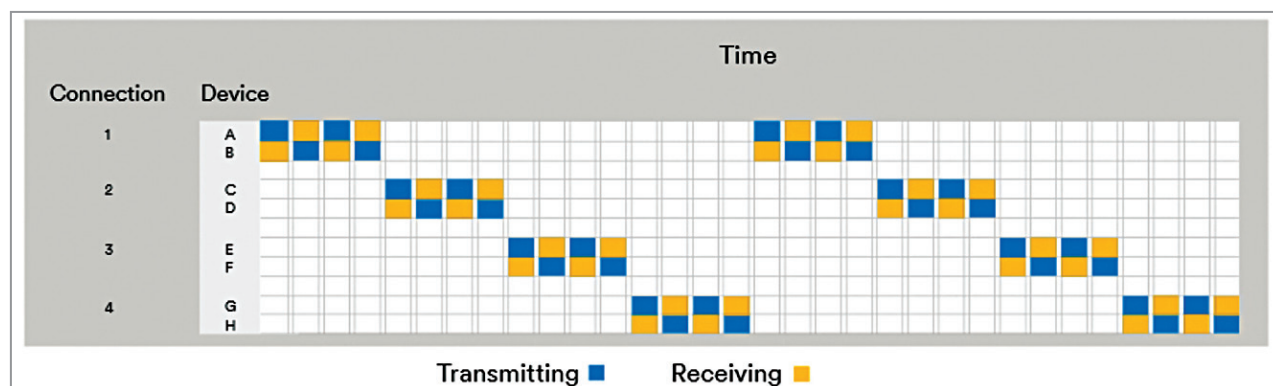


Figure 7: A simplified illustration of connection use and radio sharing

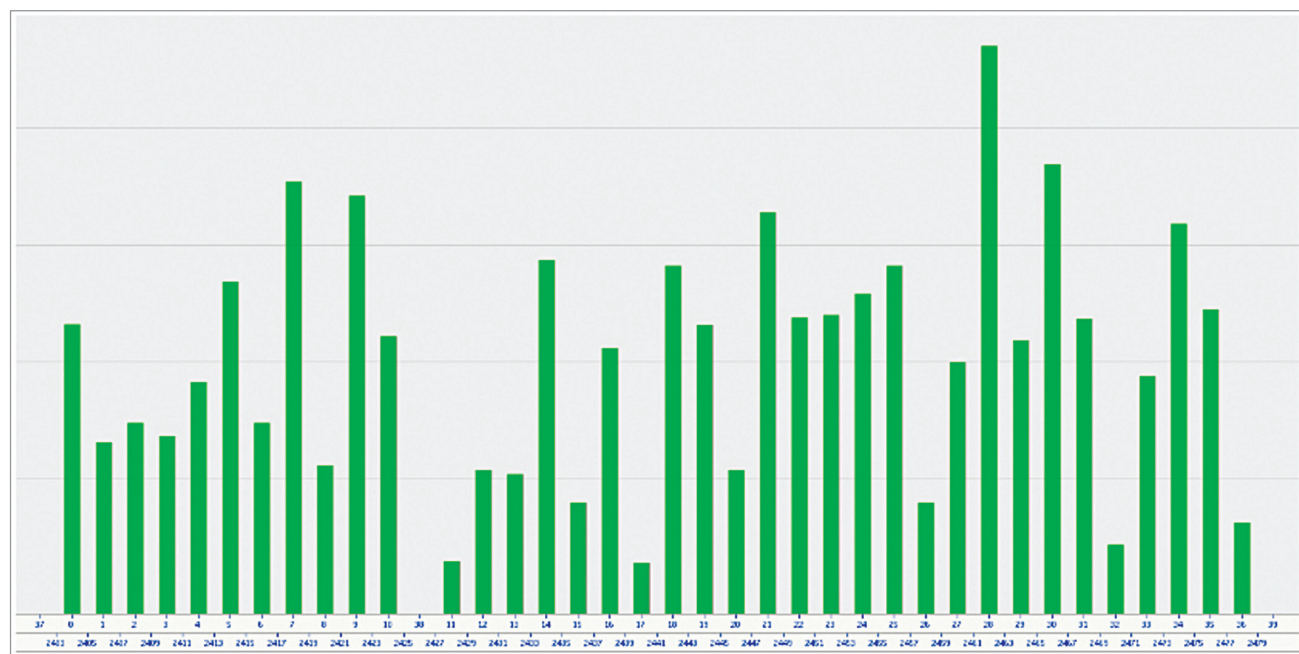


Figure 8: Adaptive frequency hopping distributing communication across channels